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$$\text{Opgave: } f(x) = \sec \frac{1}{x} = \left(\cos \frac{1}{x} \right)^{-1}$$

$$\begin{aligned} f'(x) &= \frac{\sin \frac{1}{x}}{\cos^2 \frac{1}{x}} \cdot -\frac{1}{x^2} \\ &= -\frac{\tan \frac{1}{x} \cdot \sec \frac{1}{x}}{x^2} \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{(\sec^2 \frac{1}{x} \cdot \frac{1}{x^2} \cdot \sec \frac{1}{x} + \tan \frac{1}{x} \cdot \frac{\tan \frac{1}{x} \cdot \sec \frac{1}{x}}{x^2}) \cdot x^2 - \tan \frac{1}{x} \cdot \sec \frac{1}{x} \cdot 2x}{x^4} \\ &= \frac{\sec^3 \frac{1}{x} + \tan^2 \frac{1}{x} \cdot \sec \frac{1}{x} - \tan \frac{1}{x} \cdot \sec \frac{1}{x} \cdot 2x}{x^4} \\ &= \frac{\sec \frac{1}{x} \left(\sec^2 \frac{1}{x} + \tan^2 \frac{1}{x} - \tan \frac{1}{x} \cdot 2x \right)}{x^4} \end{aligned}$$

References